Vector Databases (VDBMS)

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(Revisit) Semi-structured Big Data

- Data records have some structure, i.e., attributes and values
- The big-data framework is aware of that structure
- It utilizes this awareness in:
  - Efficient storage, e.g., Parquet column storage
  - Indexing, e.g., LSM Trees
  - Optimized query processing, e.g., Catalyst optimizer
- Can we further optimize unstructured data?

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The Rise of AI

- Unstructured queries are becoming more prevalent with the rise of Large Language Models (LLMs)
  - "Find the five cheapest 24" computer monitors that have at least 4.5 star ratings"
- Quick solution: NL2SQL

```
SELECT id, name, size_inches, price, rating
FROM monitors
WHERE size_inches = 24 AND rating >= 4.5
ORDER BY price ASC
LIMIT 5;
```


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More Interesting Unstructured Queries

- In a database of products with pictures, description, and user reviews
  - “Find highly rated products that look like this but has a white color and is preferred by graphic designers”

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AI Embedding of Records

Description

User Review

AI Model

0101011010101010

Fixed-size Embedding representing product features

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Semantic Search

"Find a highly rated product that looks like this but has a white color and is preferred by graphic designers"

AI Model

0101101011100010

Query Embedding

Nearest Neighbor Search

0101111011100011
1101101011110010
1101101001100010
0111101011100011
1101101011100010
0111101001110111

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RAG Question Answering

- RAG: Retrieval Augmented Generation
- Given a query:
  - Create a query embedding
  - Run a nearest neighbor search to find the most relevant documents
  - With an appropriate prompt, feed the documents through a generative AI model to get the final answer

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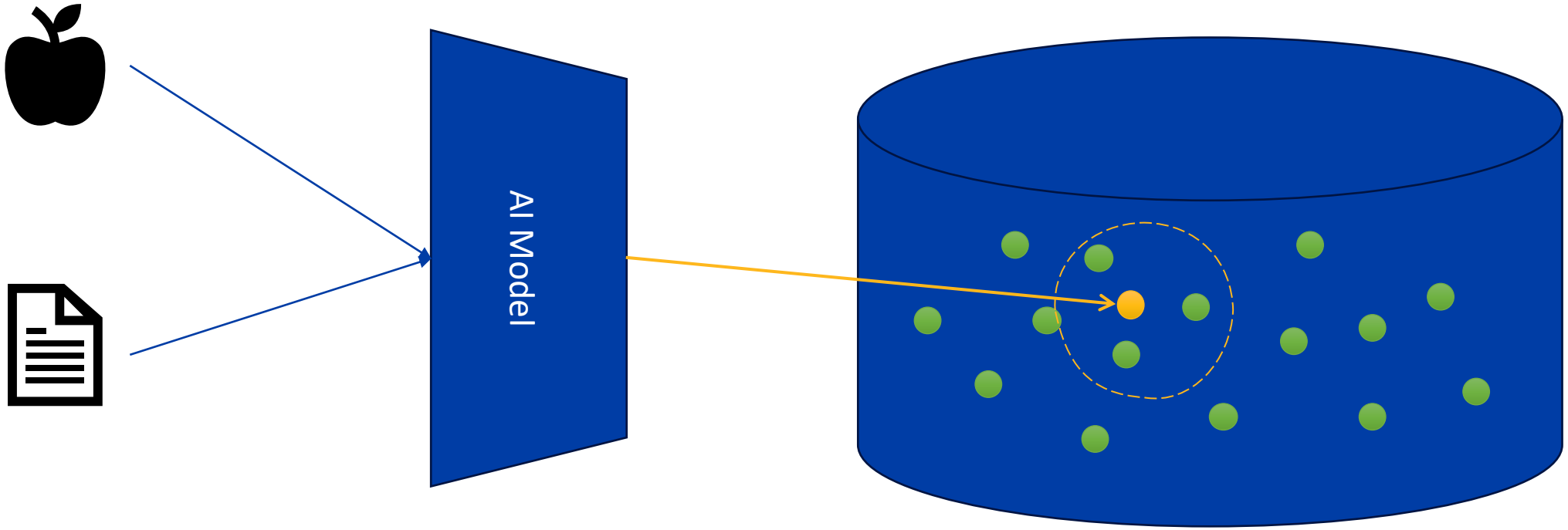
VDBMS



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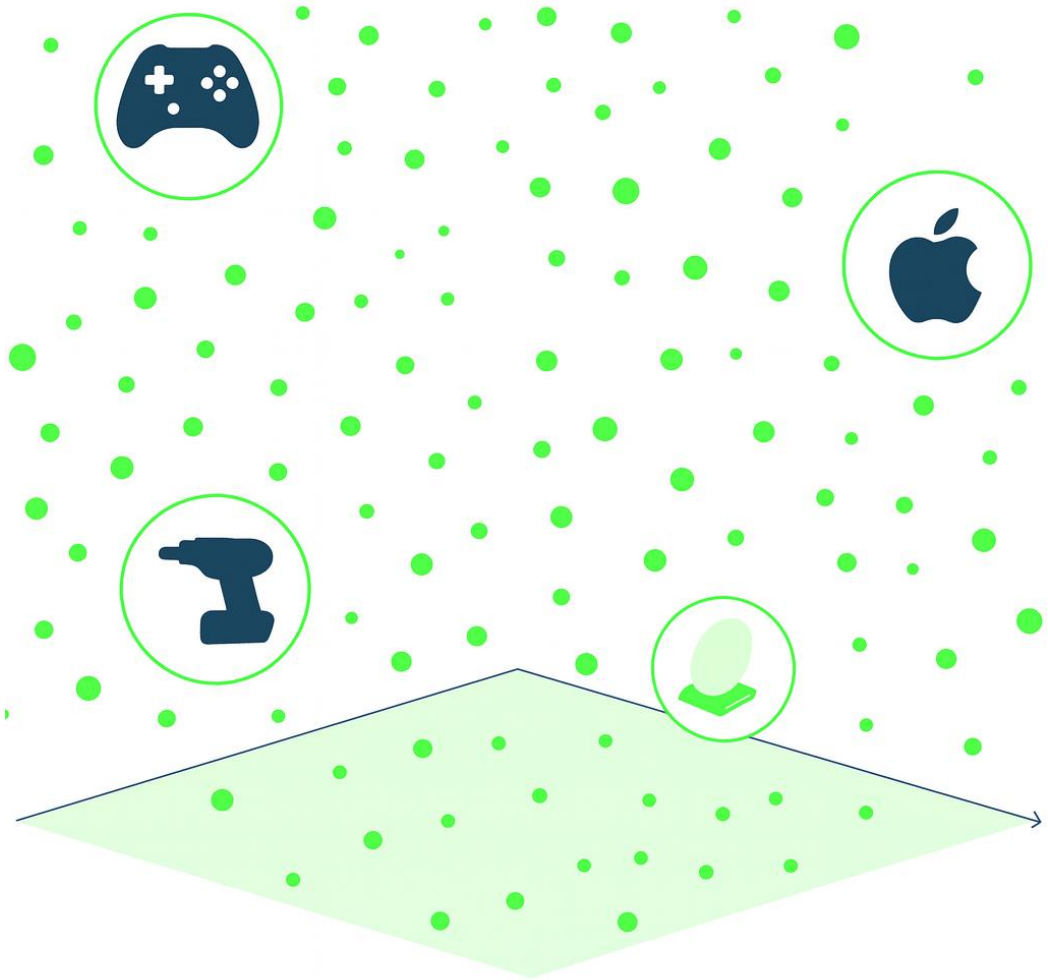
How to search vector databases



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Nearest Neighbor Search

Product ID	Title	Description	Price	Embedding
				0101111011100011
				1101101011110010
				1101101001100010

- How to answer Nearest Neighbor queries?
  - Naïve method: Full scan, sort, top-k. Inefficient
  - Index-based: Build a spatial index on embeddings and use kNN search. Curse of dimensionality
  - Approximate answers: Faster and practical

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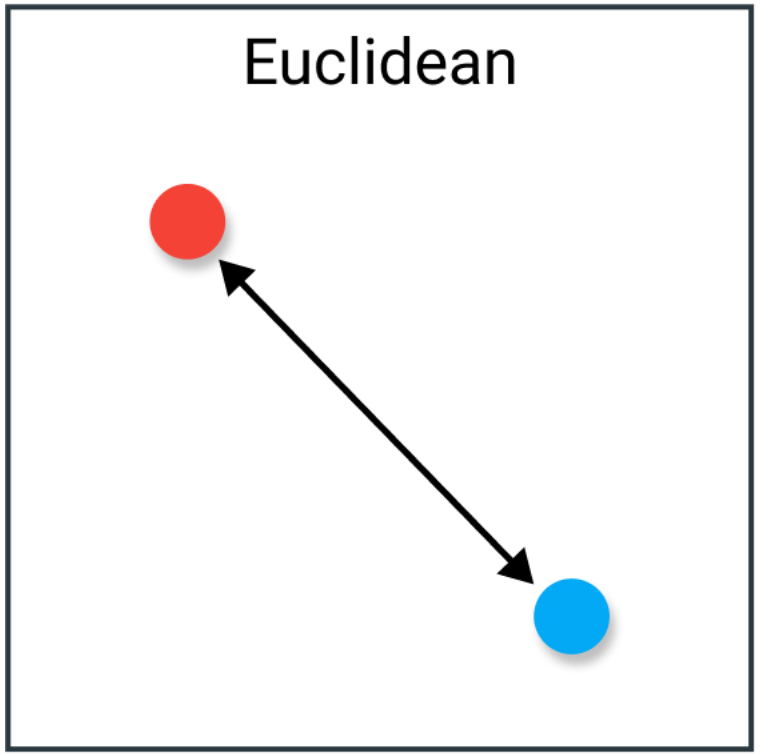
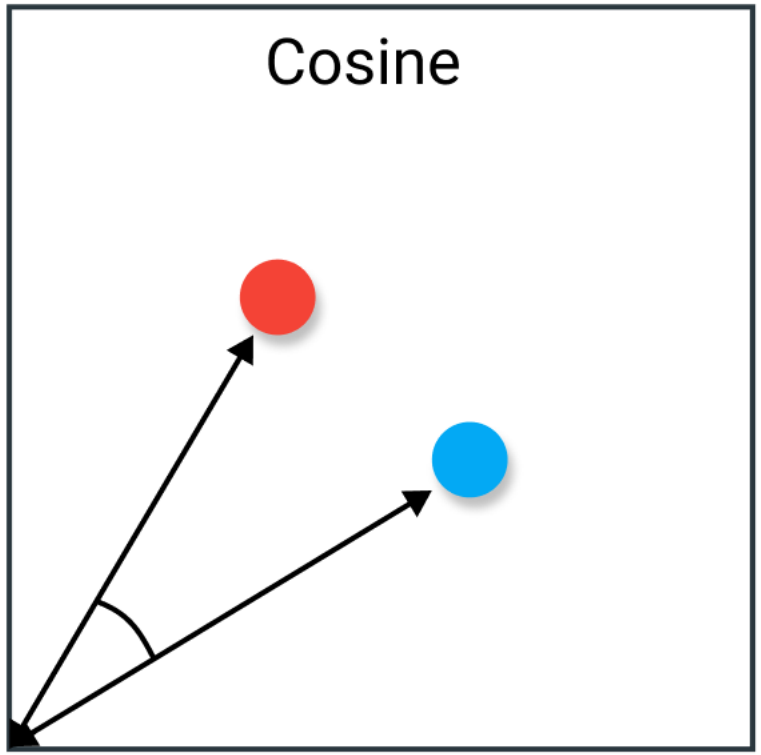
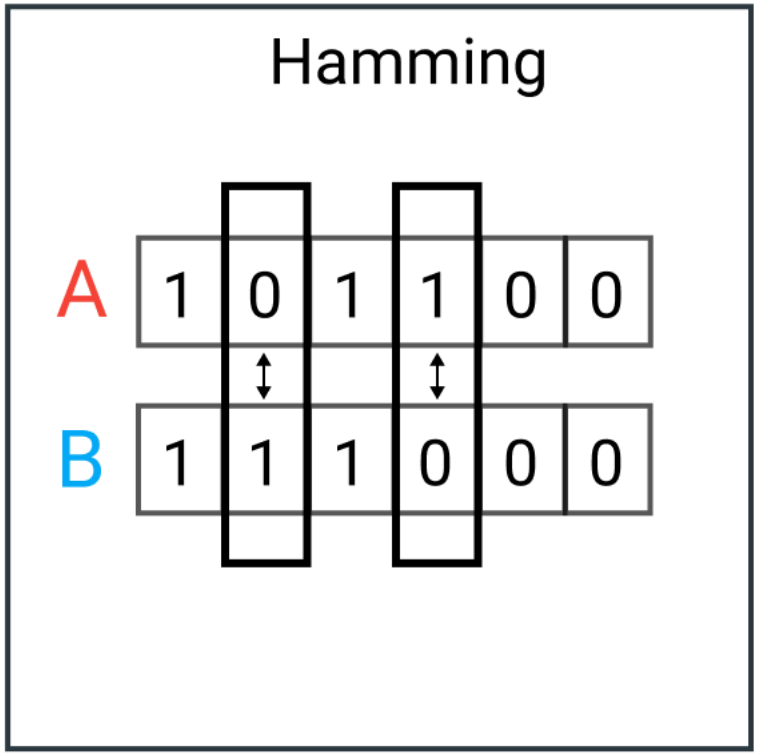
Distance Metrics







$$D(x,y) = \sqrt{\sum_{i=1}^{n}(x_i - y_i)^2)}$$

$$D(x,y) = cos(\theta) = \frac{x \cdot y}{\|x\| \, \|y\|}$$

$$D(x,y) = \sum(x_i \oplus y_i)$$

$D(x,y) = \sum(x_i \cdot y_i)$ ← Normalized vectors

Figure: https://www.maartengrootendorst.com/blog/distances/

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Vector Indexing

- A way to index multidimensional data points
- Given a query point, find the nearest data point
- Traditional spatial indexes (based on Euclidean distance) work for low-dimensional data
- As number of dimensions increases, all points become equally near

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Locality Sensitive Hashing (LSH)

- Traditional hashing is good for point queries with exact match
- Locality Sensitive Hashing (LSH) works for proximity queries

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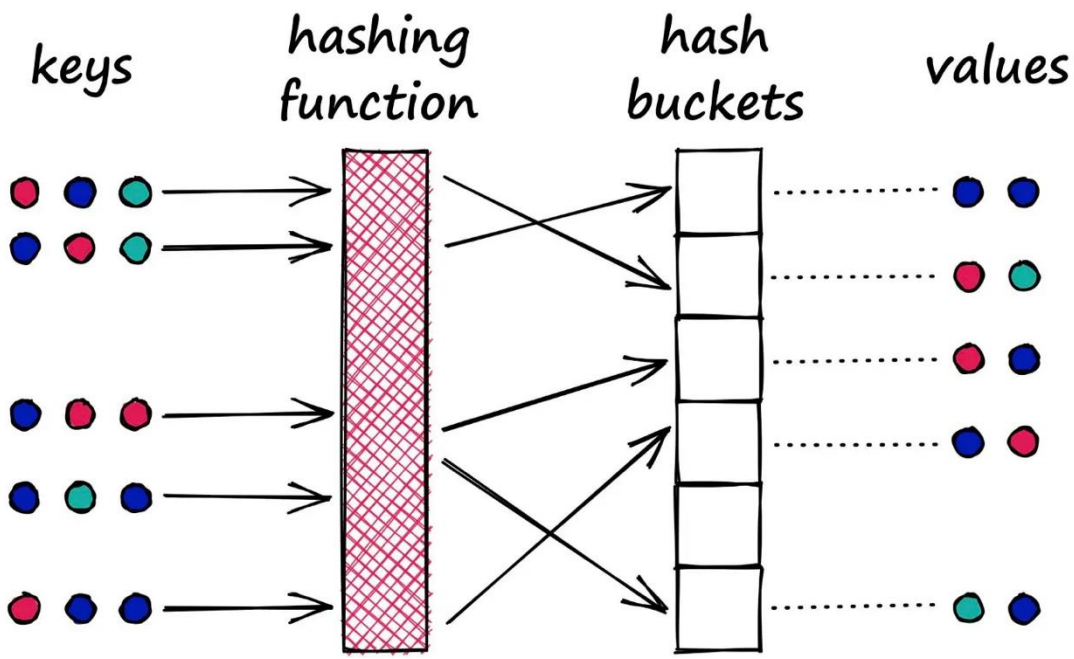
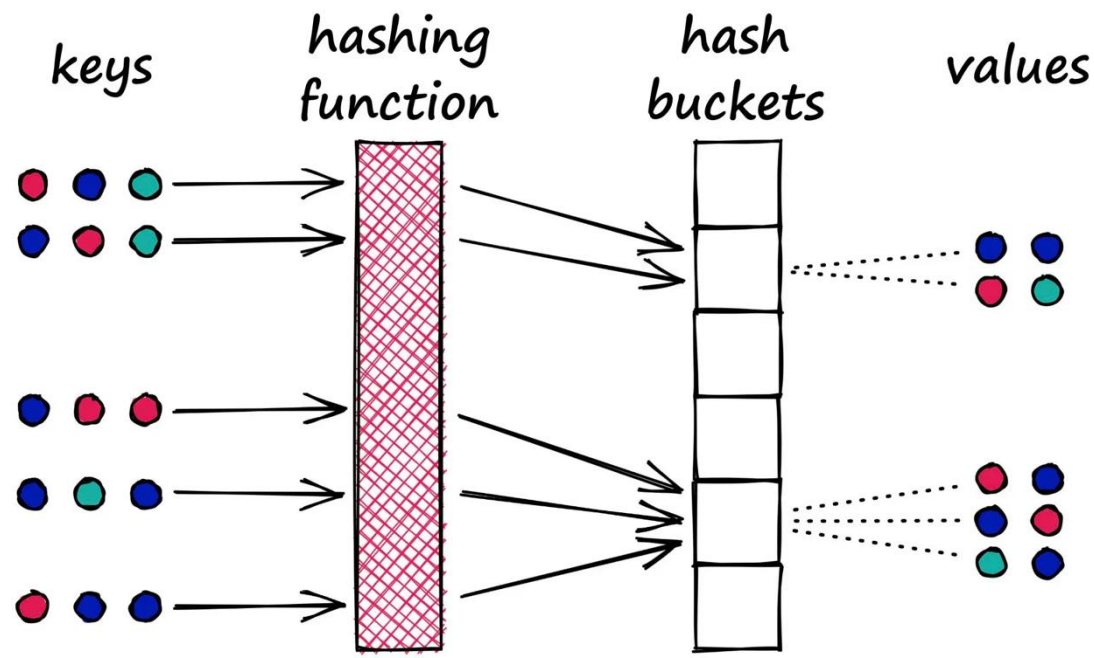


Regular Hashing



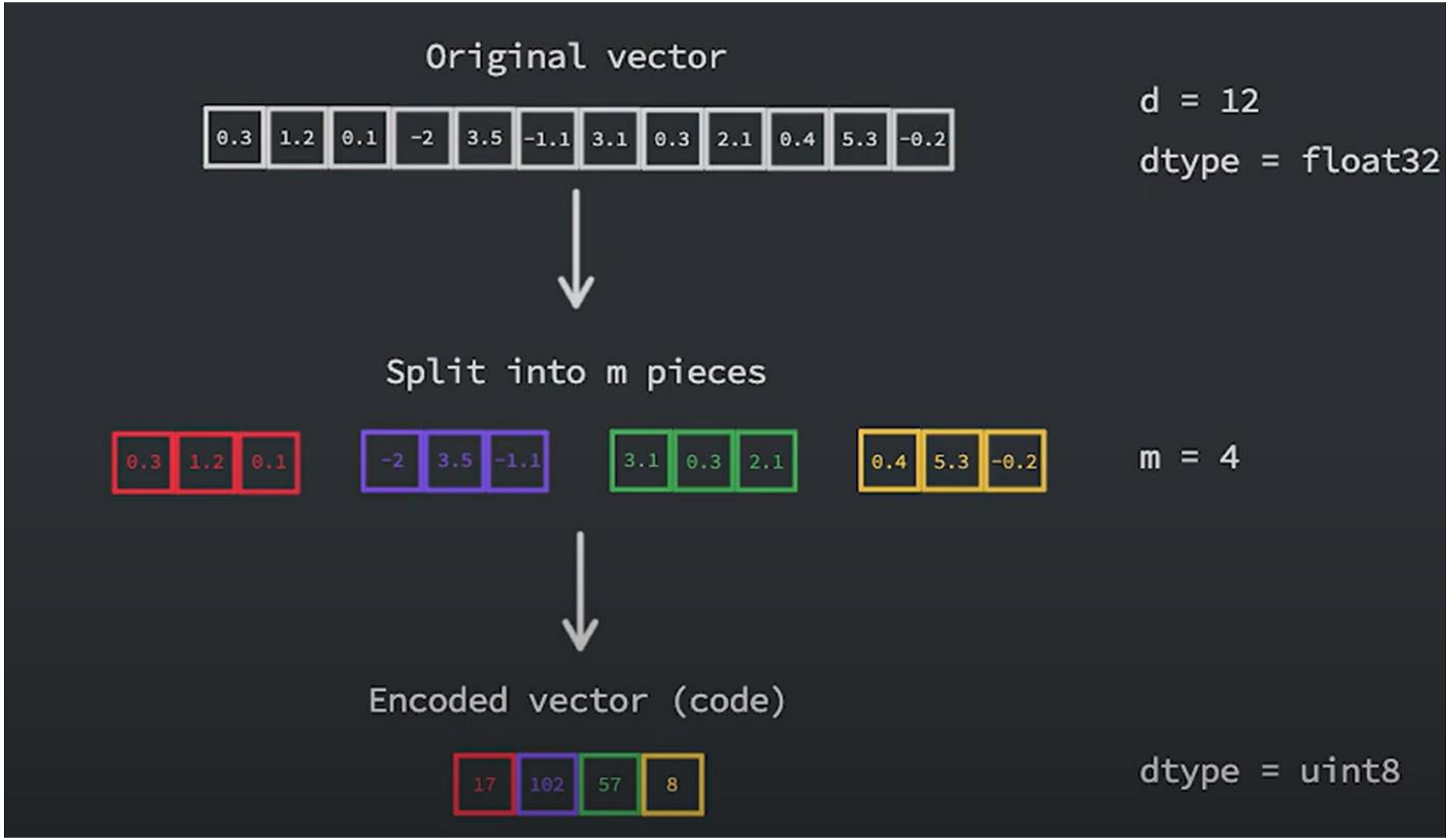
Locality Sensitive Hashing

https://www.pinecone.io/learn/series/faiss/locality-sensitive-hashing/

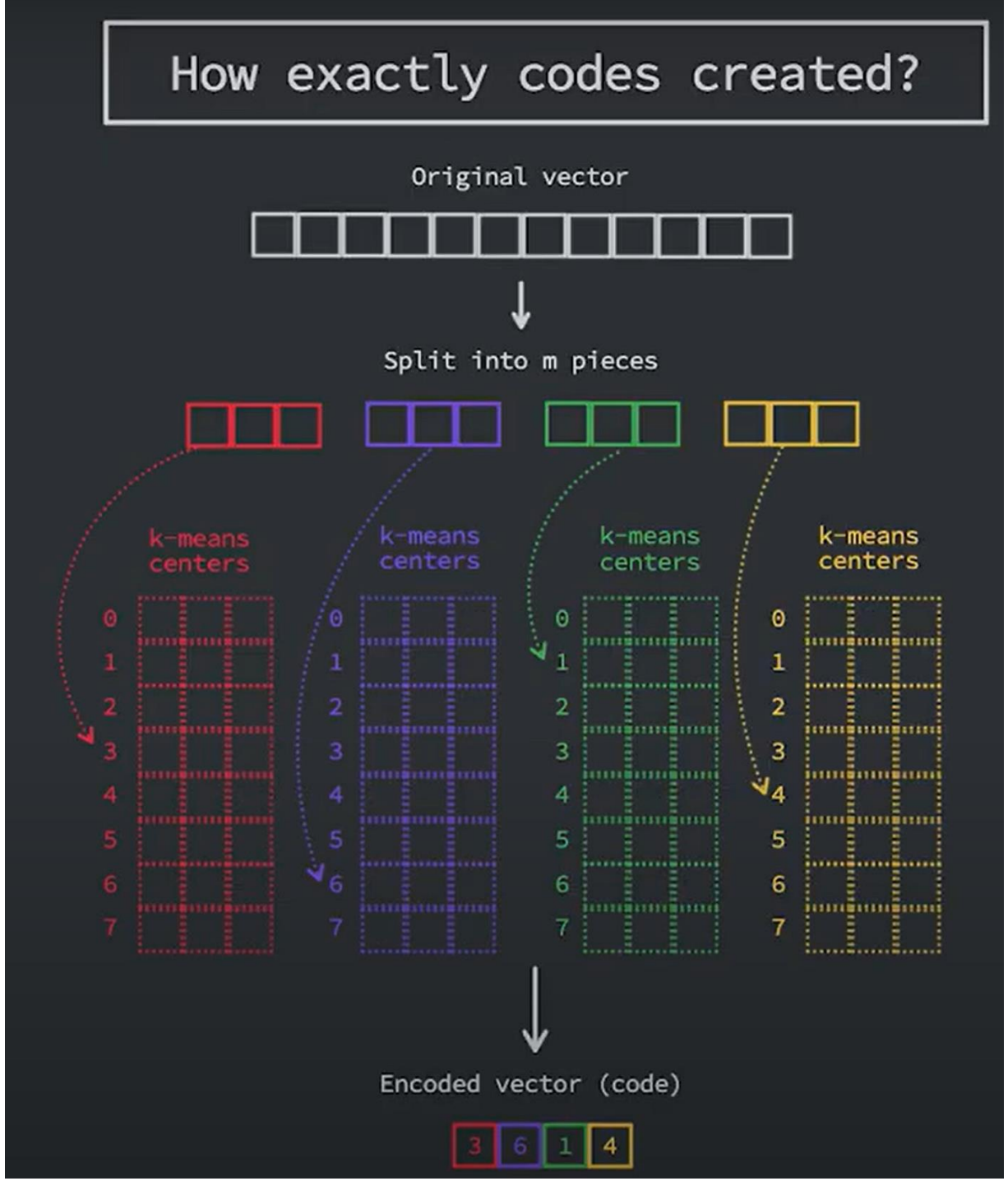
Product Quantization (PQ)

- Break down a big vector into subvectors
- Quantize the value of each subvector
  - Reduce the size of each vector
- Store a list of vector IDs (record IDs) and their corresponding quantized vector
- For a query vector (q), do the same quantization
- Compute the distances to the quantized vectors and the find the nearest one
- Examples: FAISS, Milvus, Pinecone

Product Quantization



https://medium.com/nlp-experiment/product-quantization-d66fdb860047


How exactly codes created?
Original vector
Split into m pieces
k-means centers
k-means centers
k-means centers
k-means centers
0
1
2
3
4
5
6
7
Encoded vector (code)
3
6
1
4

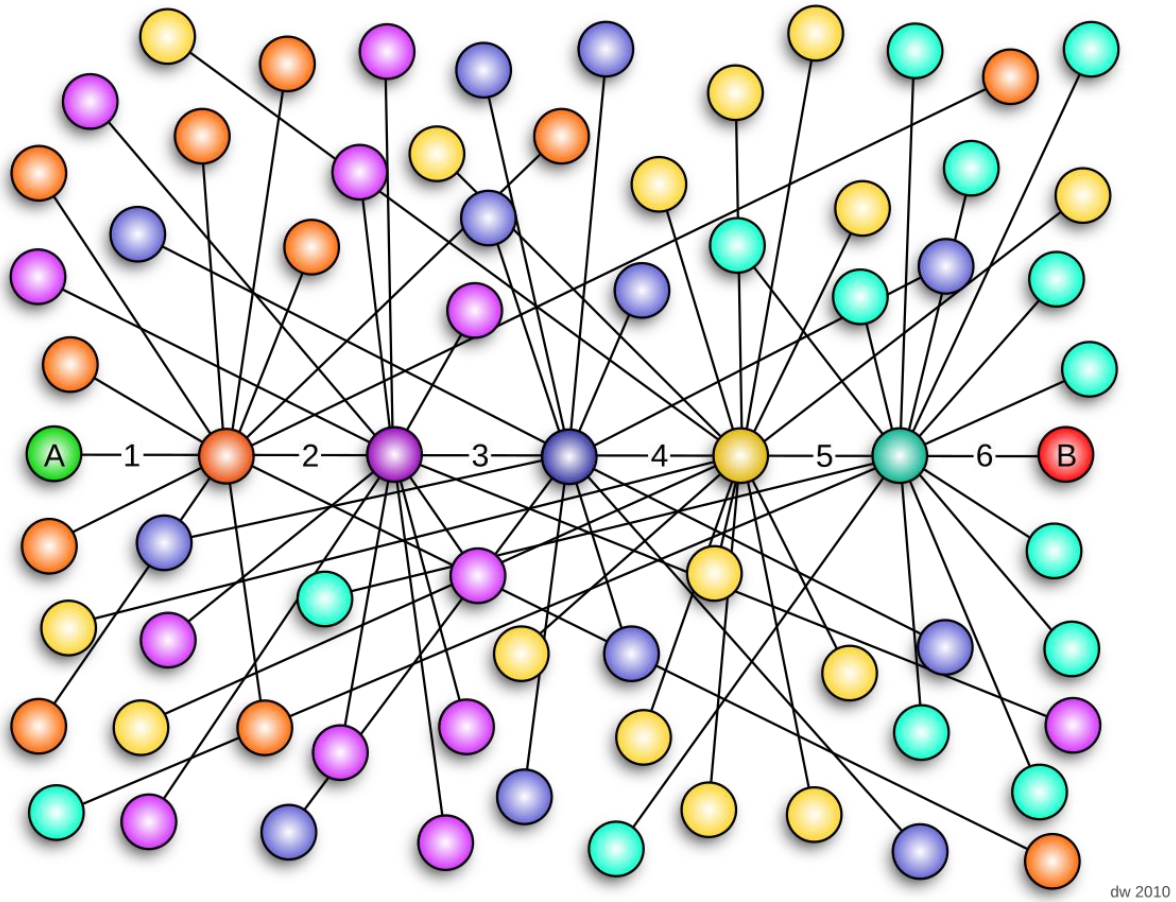
Hierarchical Navigable Small World (HNSW)

Six Degrees of Separation

- Small world: Any two people are connected with a chain of at most six people
- Milgram Experiment
  - Take a package from Nebraska and deliver it to a person in Boston
  - Use only connections you know
- Can we use it for nearest neighbor search?

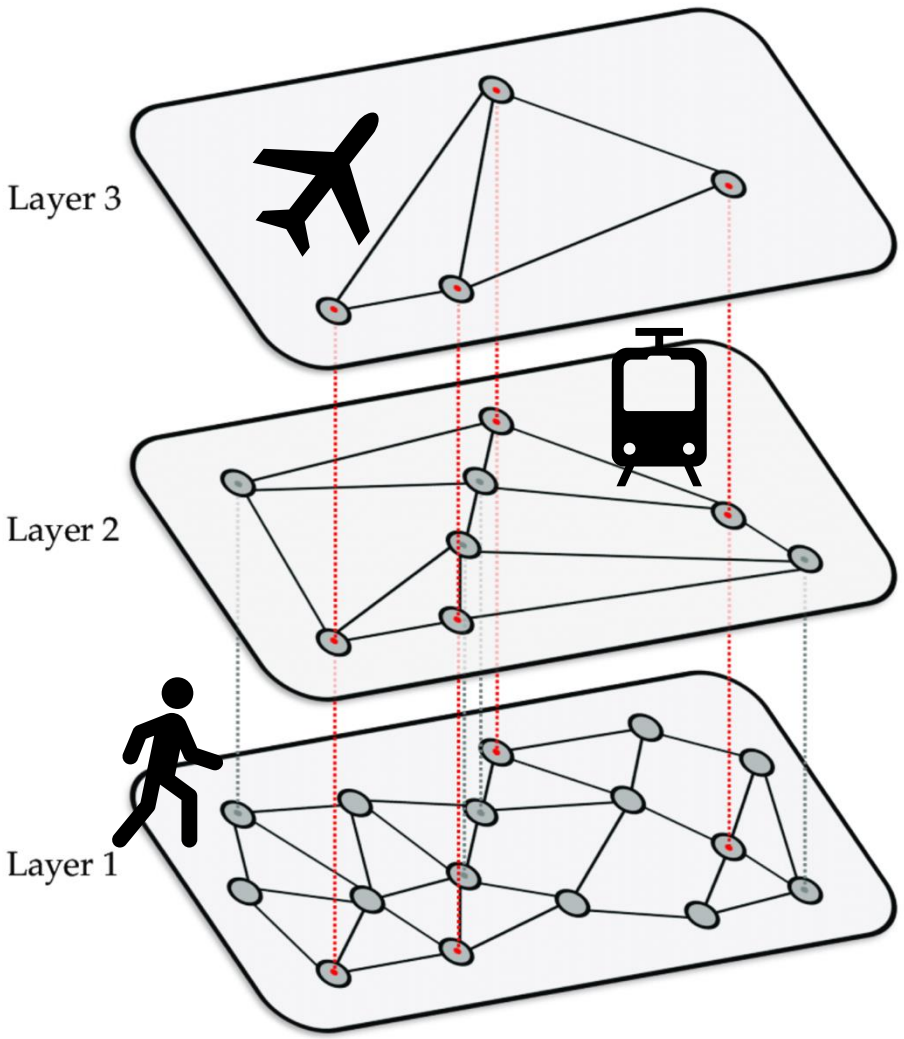
HNSW

- If you have a graph with the small-world properties, you can use it for search
- Basic concept (NSW):
  - Connect each data point to its nearest neighbors
  - Connect each data point to a few faraway neighbors
  - Search: Start with a random node and follow the edges to the target location
- Challenge: High vertex degree



HNSW

- Create a hierarchy of graphs
- Each vertex is elevated to the higher level with a probability
- Search starts at the highest level and goes to the next level when the search converges



Conclusion

- AI turns unstructured data into embeddings for semantic search.
- Querying embeddings becomes a nearest-neighbor problem.
- Exact search doesn’t scale → use Approximate NN methods.
- Key indexing tools: LSH, HNSW, Product Quantization (PQ).
- Vector DBs integrate embedding storage, ANN search, and RAG
- VDBMS Enables intelligent AI-driven applications.